Void Linux + Sway

A Comprehensive Setup Guide

A personal reference guide for building the environment from scratch on Void Linux with the Sway window manager.

Built from real-world experience and official documentation.

Each chapter is self-contained — read only what you need.

Recommended installation order:

01 → 02 → 03 → 04 → 05 → 06 → 07

Contents

Void Linux Quick Reference **5**

1 Basic Installation **7**
1.1 Update the System First . . . 7
1.2 Essential Build Tools . . . 7
1.3 GPU Drivers (Mesa without Xorg) . . . 7
1.3.1 Intel (e.g. Dell Latitude E6430) . . . 7
1.3.2 AMD . . . 7
1.3.3 NVIDIA . . . 7
1.4 Session Manager (Choose One Only!) . . . 8
1.4.1 Option A: elogind (recommended for laptops) . . . 8
1.4.2 Option B: seatd (lighter and simpler) . . . 8
1.5 XDG_RUNTIME_DIR . . . 8
1.5.1 If you chose elogind . . . 8
1.5.2 If you chose seatd . . . 8
1.6 D-Bus . . . 8
1.7 Default User Directories . . . 9
1.8 Sway and Core Tools . . . 9
1.9 XDG Desktop Portals (Required for Screen Sharing) . . . 9
1.9.1 Portal Configuration File (Important!) . . . 9
1.10 Media Codecs . . . 10
1.11 Power Management (Laptops) . . . 10
1.11.1 TLP Configuration (Important for WiFi stability) . . . 10
1.12 Fonts . . . 10
1.13 Environment Variables . . . 11
1.14 Copy the Default Config . . . 11
1.15 Essential Additions to the Sway Config . . . 12
1.16 Final Verification . . . 13

2 Audio (PipeWire + ALSA) **15**
2.1 Installation . . . 15
2.2 Configure WirePlumber . . . 15
2.3 Configure PipeWire-Pulse (for PulseAudio app support) . . . 15
2.4 Configure ALSA with PipeWire . . . 15
2.5 Starting PipeWire . . . 16
2.6 Adjusting Volume Levels . . . 16
2.7 Verification . . . 16

2.8 Common Problems . . . 16

3 Theming (Consistent Look) 19
3.1 The Basic Concept . . . 19
3.2 Installation . . . 19
3.3 Applying the Theme . . . 20
3.4 Cursor in the Sway Config . . . 20
3.5 Fonts . . . 20
3.6 Theme Locations . . . 20

4 Screenshot, Screen Recording, and Clipboard 21
4.1 Install the Tools . . . 21
4.2 Screenshots . . . 21
4.2.1 Direct Commands . . . 21
4.2.2 In the Sway Config . . . 21
4.3 Screen Recording with wf-recorder . . . 21
4.4 Clipboard Manager with clipman . . . 22
4.4.1 Start (in Sway config) . . . 22
4.4.2 Browse History . . . 22
4.4.3 Clear History . . . 22

5 Hardware (Bluetooth, Printer, USB, MTP) 23
5.1 Bluetooth . . . 23
5.1.1 Installation . . . 23
5.1.2 Usage (CLI) . . . 23
5.1.3 Verification . . . 23
5.2 Printer with CUPS . . . 23
5.2.1 Installation . . . 23
5.2.2 Adding a Printer . . . 24
5.3 USB Auto-mount with udiskie . . . 24
5.3.1 Installation . . . 24
5.3.2 Polkit Rule (to avoid password prompts) . . . 24
5.3.3 Manual Unmount . . . 24
5.4 MTP (Phone via USB) . . . 24
5.4.1 FUSE Setup . . . 25
5.4.2 Usage . . . 25

6 Networking (VPN, Hotspot, Tethering) 27
6.1 NetworkManager Setup . . . 27
6.1.1 Static DNS (optional) . . . 27
6.2 VPN . . . 27
6.2.1 Install Plugins . . . 27
6.2.2 Add a VPN Connection . . . 28
6.2.3 Enable / Disable . . . 28
6.3 Hotspot . . . 28
6.3.1 Quick Start . . . 28
6.3.2 Show QR Code . . . 28
6.3.3 Persistent Hotspot . . . 28
6.4 USB Tethering (Phone Internet on Laptop) . . . 29

7 Applications **31**
7.1 Browser . . . 31
7.2 Multimedia . . . 31
7.3 File Manager . . . 31
7.4 Archive Tools . . . 31
7.5 System Monitoring . . . 32
7.6 Flatpak . . . 32
7.7 AppImage . . . 32
7.8 Emacs . . . 32
7.9 Terminal Tools for Coding . . . 32
7.9.1 Link fzf to Shell History . . . 33

8 AppImage and Desktop Entries **35**
8.1 Directory Structure . . . 35
8.2 Setting Up a New AppImage . . . 35
8.3 Desktop Entry Template . . . 35
8.4 Exec Variables . . . 36
8.5 Common MIME Types . . . 36
8.5.1 Useful Commands . . . 36
8.6 Full Example: Obsidian . . . 36
8.7 Web App as a Desktop Entry . . . 37
8.8 Quick Reference . . . 37

9 KVM/QEMU Virtualization **39**
9.1 Verify CPU Virtualization Support . . . 39
9.2 Installation . . . 39
9.3 Enable Services . . . 39
9.4 Add User to the Group . . . 40
9.5 Load the KVM Module . . . 40
9.6 Configure libvirt (to run VMs without sudo) . . . 40
9.7 Post-Reboot Verification . . . 40
9.8 Essential virsh Commands . . . 41
9.9 Nested Virtualization . . . 41
9.10 Troubleshooting . . . 41

10 Troubleshooting **43**
10.1 Sway Won't Start . . . 43
10.1.1 "Failed to create backend" . . . 43
10.1.2 Sway starts and crashes immediately . . . 43
10.2 XDG_RUNTIME_DIR Issues . . . 43
10.3 Audio . . . 44
10.3.1 PipeWire not running . . . 44
10.4 Firefox Not Running on Wayland . . . 44
10.5 Screen Sharing Not Working . . . 44
10.6 Qt Apps Showing as X11 . . . 44
10.7 Java Apps (Blank Grey Window) . . . 45
10.8 AppImage Won't Launch . . . 45
10.9 General Diagnostic Commands . . . 45

11 xbps-src: Building Packages Not in the Repo **47**
11.1 The Core Concept 47
11.2 Initial Setup (One Time Only) 47
11.3 Building a Package Already in void-packages 47
11.3.1 Practical Example: Google Chrome 48
11.4 Adding the Local Repo Permanently 48
11.5 Anatomy of a Template 48
11.5.1 Key Variables 49
11.5.2 build_style Options 49
11.6 Getting the Checksum 49
11.7 Full Example: A Rust Package (zoxide) 49
11.8 Everyday xbps-src Commands 50
11.9 Updating a Package 50
11.10 Hold: Prevent a Package from Updating 50
11.11 Troubleshooting xbps-src 50
11.12 Quick Summary 51

12 i3 on Void Linux (Migration from Sway + Running Both) **53**
12.1 The Core Difference in One Sentence 53
12.2 Full Migration Table: Swap These for Those 54
12.3 Installing i3 on Void 54
12.4 Launching i3 from the TTY 55
12.5 Auto-start from TTY1 (Sway) or TTY2 (i3) 55
12.6 i3 Config (A Full Starting Point) 56
12.7 Sharing Config Between i3 and Sway (The Smart Way) 57
12.7.1 Structure 57
12.7.2 Implementation 57
12.8 Theming in i3 58
12.9 picom (Compositor for i3) 58
12.10 Summary: What to Install on Top of What You Have 58
12.11 Quick Verification 59

Void Linux Quick Reference

```
# Install a package
sudo xbps-install PACKAGE

# Enable a service
sudo ln -s /etc/sv/SERVICENAME /var/service/

# After any "usermod -aG" --> log out and back in

# Launch Sway correctly
dbus-run-session sway

# elogind vs seatd --> pick ONE, they cannot run together
```

Chapter 1

Basic Installation

1.1 Update the System First

```
sudo xbps-install -Su
```

1.2 Essential Build Tools

```
sudo xbps-install base-devel git make ca-certificates wget
   curl
```

1.3 GPU Drivers (Mesa without Xorg)

On Wayland you do **not** need `xorg-server` or `xf86-video-intel`.

1.3.1 Intel (e.g. Dell Latitude E6430)

```
sudo xbps-install mesa-dri intel-video-accel \
  libva-utils mesa-vaapi mesa-vdpau \
  vulkan-loader mesa-vulkan-intel
```

1.3.2 AMD

```
sudo xbps-install mesa-dri mesa-vaapi mesa-vdpau \
  vulkan-loader mesa-vulkan-radeon
```

1.3.3 NVIDIA

```
sudo xbps-install nvidia nvidia-libs
```

1.4 Session Manager (Choose One Only!)

Warning: `elogind` and `seatd` cannot run together — choose one.

1.4.1 Option A: elogind (recommended for laptops)

```
sudo xbps-install elogind
sudo ln -s /etc/sv/elogind /var/service/
```

elogind advantages:

- Creates `XDG_RUNTIME_DIR` automatically
- Supports power management (suspend, hibernate)
- More stable with most applications

1.4.2 Option B: seatd (lighter and simpler)

```
sudo xbps-install seatd
sudo ln -s /etc/sv/seatd /var/service/
sudo usermod -aG _seatd $USER
```

seatd advantages:

- Much lighter
- Manages only the seat (not `XDG_RUNTIME_DIR`)
- Requires `pam_rundir` alongside it

1.5 XDG_RUNTIME_DIR

1.5.1 If you chose elogind

Nothing needed — it is created automatically.

1.5.2 If you chose seatd

```
sudo xbps-install pam_rundir
```

Add to `/etc/pam.d/system-login`:

```
-session optional pam_rundir.so
```

1.6 D-Bus

```
sudo xbps-install dbus
sudo ln -s /etc/sv/dbus /var/service/
```

1.7 Default User Directories

```
sudo xbps-install xdg-user-dirs
xdg-user-dirs-update
```

1.8 Sway and Core Tools

```
sudo xbps-install \
  sway swaybg swayidle swaylock \
  xorg-server-xwayland \
  wlroots wl-clipboard

# Terminal
sudo xbps-install tilix   # or foot (lighter and faster)
sudo xbps-install foot

# App Launcher
sudo xbps-install fuzzel   # or wofi
sudo xbps-install wofi

# Status Bar
sudo xbps-install waybar

# Notifications
sudo xbps-install mako libnotify

# Screenshot
sudo xbps-install grim slurp

# Screen Recording
sudo xbps-install wf-recorder

# Clipboard Manager
sudo xbps-install clipman
```

1.9 XDG Desktop Portals (Required for Screen Sharing)

```
sudo xbps-install \
  xdg-desktop-portal \
  xdg-desktop-portal-wlr \
  xdg-desktop-portal-gtk \
  xdg-utils
```

1.9.1 Portal Configuration File (Important!)

```
mkdir -p ~/.config/xdg-desktop-portal
```

```
nano ~/.config/xdg-desktop-portal/sway-portals.conf
```

```
[preferred]
default=gtk
org.freedesktop.impl.portal.Screenshot=wlr
org.freedesktop.impl.portal.ScreenCast=wlr
```

The file must be named `sway-portals.conf` exactly, because `XDG_CURRENT_DESKTOP=sway`.

1.10 Media Codecs

```
sudo xbps-install \
  ffmpeg \
  gst-plugins-base1 \
  gst-plugins-good1 \
  gst-plugins-bad1 \
  gst-plugins-ugly1
```

1.11 Power Management (Laptops)

```
sudo xbps-install tlp acpid acpi brightnessctl
sudo ln -s /etc/sv/acpid /var/service/
sudo ln -s /etc/sv/tlpd /var/service/
sudo usermod -aG video $USER
```

1.11.1 TLP Configuration (Important for WiFi stability)

```
sudo nano /etc/tlp.conf
```

Find and change:

```
WIFI_PWR_ON_AC=off
WIFI_PWR_ON_BAT=off
```

1.12 Fonts

```
sudo xbps-install \
  liberation-fonts-ttf \
  dejavu-fonts-ttf \
  noto-fonts-ttf \
  noto-fonts-ttf-extra \
  noto-fonts-emoji \
  font-ibm-plex-ttf \
  terminus-font \
  font-awesome6
```

JetBrains Mono: Not in the repo — download it manually from https://www.jetbrains.com/lp/mono/ and place it in `~/.local/share/fonts/`

1.13 Environment Variables

Add to `~/.bash_profile` or `~/.profile`:

```
# XDG Base Directories
export XDG_CONFIG_HOME="$HOME/.config"
export XDG_DATA_HOME="$HOME/.local/share"
export XDG_CACHE_HOME="$HOME/.cache"
export XDG_STATE_HOME="$HOME/.local/state"
export PATH="$HOME/.local/bin:$PATH"

# XDG_RUNTIME_DIR (if not using elogind)
if test -z "${XDG_RUNTIME_DIR}"; then
    export XDG_RUNTIME_DIR=/tmp/${UID}-runtime-dir
    if ! test -d "${XDG_RUNTIME_DIR}"; then
        mkdir "${XDG_RUNTIME_DIR}"
        chmod 0700 "${XDG_RUNTIME_DIR}"
    fi
fi

# Wayland
export XDG_SESSION_TYPE=wayland
export XDG_SESSION_DESKTOP=sway
export XDG_CURRENT_DESKTOP=sway
export DE=sway

# Application hints
export MOZ_ENABLE_WAYLAND=1
export QT_QPA_PLATFORM=wayland
export QT_WAYLAND_DISABLE_WINDOWDECORATION=1
export SDL_VIDEODRIVER=wayland
export ELECTRON_OZONE_PLATFORM_HINT=auto
export _JAVA_AWT_WM_NONREPARENTING=1

# If using seatd
export LIBSEAT_BACKEND=seatd

# Auto-start Sway from tty1
if [ -z "$DISPLAY" ] && [ -z "$WAYLAND_DISPLAY" ] && [
   "${XDG_VTNR:-0}" -eq 1 ]; then
    exec dbus-run-session sway
fi
```

1.14 Copy the Default Config

```
mkdir -p ~/.config/sway
```

```
cp /etc/sway/config ~/.config/sway/config
```

1.15 Essential Additions to the Sway Config

Open `~/.config/sway/config` and add:

```
### Update dbus environment ###
exec dbus-update-activation-environment DISPLAY
   WAYLAND_DISPLAY SWAYSOCK XDG_CURRENT_DESKTOP

### XDG Desktop Portal ###
exec /usr/libexec/xdg-desktop-portal-wlr &
exec /usr/libexec/xdg-desktop-portal -r &

### PipeWire ###
exec pipewire &

### Notifications ###
exec mako &

### Polkit ###
exec /usr/libexec/polkit-gnome-authentication-agent-1 &

### Network ###
exec nm-applet --indicator &

### Bluetooth ###
exec blueman-applet &

### Clipboard ###
exec wl-paste -t text --watch clipman store &

### USB Auto-mount ###
exec udiskie --tray &

### Keyboard Layout ###
input type:keyboard {
    xkb_layout us,ara
    xkb_options grp:lwin_toggle
    repeat_delay 300
    repeat_rate 50
}

### Touchpad ###
input type:touchpad {
    tap enabled
    natural_scroll enabled
    dwt enabled
}
```

```
### Key Bindings ###
# Screenshot
bindsym Print exec grim ~/Pictures/screenshot-$(date
   +%Y%m%d-%H%M%S).png
bindsym Shift+Print exec grim -g "$(slurp)"
   ~/Pictures/screenshot-$(date +%Y%m%d-%H%M%S).png
bindsym Ctrl+Print exec grim -g "$(slurp)" - | wl-copy

# Volume
bindsym XF86AudioRaiseVolume exec pactl set-sink-volume
   @DEFAULT_SINK@ +5%
bindsym XF86AudioLowerVolume exec pactl set-sink-volume
   @DEFAULT_SINK@ -5%
bindsym XF86AudioMute exec pactl set-sink-mute
   @DEFAULT_SINK@ toggle

# Brightness
bindsym XF86MonBrightnessUp exec brightnessctl set +10%
bindsym XF86MonBrightnessDown exec brightnessctl set 10%-

# Lock screen
bindsym $mod+l exec swaylock -f -c 000000

### Idle + Lock ###
exec swayidle -w \
    timeout 300 'swaylock -f -c 000000' \
    timeout 600 'swaymsg "output * dpms off"' \
    resume 'swaymsg "output * dpms on"' \
    before-sleep 'swaylock -f -c 000000'
```

1.16 Final Verification

```
sv status dbus
sv status elogind   # or seatd
groups $USER
echo $XDG_RUNTIME_DIR
echo $WAYLAND_DISPLAY   # after launching Sway: wayland-1
pactl info | grep "Server Name"
grim ~/test.png
```

Chapter 2

Audio (PipeWire + ALSA)

2.1 Installation

```
sudo xbps-install \
  pipewire \
  wireplumber \
  alsa-utils \
  alsa-tools \
  alsa-pipewire \
  libspa-bluetooth \
  pavucontrol
```

`libspa-bluetooth` is required for Bluetooth headphones to work with PipeWire.

2.2 Configure WirePlumber

```
mkdir -p ~/.config/pipewire/pipewire.conf.d

ln -s /usr/share/examples/wireplumber/10-wireplumber.conf \
      ~/.config/pipewire/pipewire.conf.d/
```

2.3 Configure PipeWire-Pulse (for PulseAudio app support)

```
sudo xbps-install pipewire-pulse

ln -s /usr/share/examples/pipewire/20-pipewire-pulse.conf \
      ~/.config/pipewire/pipewire.conf.d/
```

2.4 Configure ALSA with PipeWire

```
sudo mkdir -p /etc/alsa/conf.d

sudo ln -s /usr/share/alsa/alsa.conf.d/50-pipewire.conf \
           /etc/alsa/conf.d/

sudo ln -s
   /usr/share/alsa/alsa.conf.d/99-pipewire-default.conf \
           /etc/alsa/conf.d/

alsactl init
```

2.5 Starting PipeWire

On Void, PipeWire is **started from inside Sway**, not as a system service.

In your Sway config:

```
exec pipewire &
```

Warning: Do not try to enable it with `ln -s /etc/sv/pipewire /var/service/` — it will not work correctly.

2.6 Adjusting Volume Levels

```
# Command line
alsamixer

# Graphical
pavucontrol
```

2.7 Verification

```
pactl info | grep "Server Name"
# Expected: PulseAudio (on PipeWire ...)

pactl list sources short
pactl list sinks short
```

2.8 Common Problems

No audio:

```
pactl info
pipewire &
pactl info
```

Bluetooth headphones not working:

```
sudo xbps-install libspa-bluetooth
# Then restart Sway
```

No sound device found:

```
aplay -l
# If empty, check GPU drivers and kernel modules
```

Chapter 3

Theming (Consistent Look)

3.1 The Basic Concept

A Wayland theme consists of 4 separate layers:

Layer	Effect
GTK Theme	Colors and shape of buttons and windows
Icon Theme	File and application icons
Cursor Theme	Mouse pointer appearance
Font	Text font in all applications

The problem: On Wayland, GTK does not read settings automatically — they must be set via `gsettings`. **The solution:** Use the `nwg-look` tool.

3.2 Installation

```
sudo xbps-install nwg-look

# Materia (flat, modern)
sudo xbps-install materia-theme

# Arc
sudo xbps-install arc-theme

# Adwaita:dark -- included with GTK by default
```

Download a theme from the internet (e.g. Catppuccin):

```
mkdir -p ~/.local/share/themes
cd ~/.local/share/themes
# Extract the downloaded zip here
```

```
# Icon Theme
sudo xbps-install papirus-icon-theme
```

```
# Cursor Theme
sudo xbps-install breeze-cursors
```

3.3 Applying the Theme

```
nwg-look
# Choose GTK Theme + Icons + Cursor + Font
# Click Apply
```

3.4 Cursor in the Sway Config

Add to `~/.config/sway/config`:

```
seat seat0 xcursor_theme THEME_NAME 24
```

3.5 Fonts

```
sudo xbps-install \
  font-jetbrains-mono \
  noto-fonts-ttf \
  noto-fonts-ttf-extra \
  noto-fonts-emoji \
  font-awesome6 \
  dejavu-fonts-ttf

# Rebuild cache after installing new fonts
fc-cache -fv
```

In the Sway config:

```
font pango:JetBrains Mono 11, Font Awesome 6 Free 11
```

3.6 Theme Locations

```
# GTK themes
/usr/share/themes/
~/.local/share/themes/

# Icons + Cursors
/usr/share/icons/
~/.local/share/icons/

# List installed themes
ls /usr/share/themes/
ls /usr/share/icons/
fc-list | grep -i "jet\|noto"
```

Chapter 4

Screenshot, Screen Recording, and Clipboard

4.1 Install the Tools

```
sudo xbps-install grim slurp wf-recorder wl-clipboard
   clipman
```

4.2 Screenshots

4.2.1 Direct Commands

```
# Full screen
grim ~/Pictures/screenshot-$(date +%Y%m%d-%H%M%S).png

# Selected region
grim -g "$(slurp)" ~/Pictures/screenshot-$(date
   +%Y%m%d-%H%M%S).png

# Copy directly to clipboard
grim -g "$(slurp)" - | wl-copy
```

4.2.2 In the Sway Config

```
bindsym Print exec grim ~/Pictures/screenshot-$(date
   +%Y%m%d-%H%M%S).png
bindsym Shift+Print exec grim -g "$(slurp)"
   ~/Pictures/screenshot-$(date +%Y%m%d-%H%M%S).png
bindsym Ctrl+Print exec grim -g "$(slurp)" - | wl-copy
```

4.3 Screen Recording with wf-recorder

```
# Full screen
wf-recorder -f ~/Videos/recording.mp4

# Selected region
wf-recorder -g "$(slurp)" -f ~/Videos/recording.mp4

# With audio
wf-recorder -a -f ~/Videos/recording.mp4

# With a specific audio device
pactl list sources short          # list available devices
wf-recorder -a "source_name" -f ~/Videos/recording.mp4

# Stop recording
# Ctrl+C
```

4.4 Clipboard Manager with clipman

4.4.1 Start (in Sway config)

```
exec wl-paste -t text --watch clipman store &
```

4.4.2 Browse History

```
clipman pick --tool wofi
# or
clipman pick --tool fuzzel
```

Add to Sway config:

```
bindsym $mod+shift+v exec clipman pick --tool fuzzel
```

4.4.3 Clear History

```
clipman clear --all
```

Chapter 5

Hardware (Bluetooth, Printer, USB, MTP)

5.1 Bluetooth

5.1.1 Installation

```
sudo xbps-install bluez blueman libspa-bluetooth
sudo ln -s /etc/sv/bluetoothd /var/service/
sudo usermod -aG bluetooth $USER
# Log out and back in
```

In the Sway config:

```
exec blueman-applet &
```

5.1.2 Usage (CLI)

```
bluetoothctl
power on
scan on
pair XX:XX:XX:XX:XX:XX
connect XX:XX:XX:XX:XX:XX
trust XX:XX:XX:XX:XX:XX
```

5.1.3 Verification

```
sv status bluetoothd
```

5.2 Printer with CUPS

5.2.1 Installation

```
sudo xbps-install cups cups-filters gutenprint
   system-config-printer
# HP printers:
sudo xbps-install hplip
# Epson printers:
sudo xbps-install epson-inkjet-printer-escpr

sudo ln -s /etc/sv/cupsd /var/service/
sudo usermod -aG lpadmin $USER
```

5.2.2 Adding a Printer

Open in your browser: `http://localhost:631`
Then: Administration → Add Printer

5.3 USB Auto-mount with udiskie

5.3.1 Installation

```
sudo xbps-install udisks2 udiskie gvfs ntfs-3g exfat-utils
```

5.3.2 Polkit Rule (to avoid password prompts)

```
sudo mkdir -p /etc/polkit-1/rules.d/
sudo nano /etc/polkit-1/rules.d/50-udisks.rules
```

```
polkit.addRule(function(action, subject) {
  if (subject.isInGroup("wheel")) {
    if (action.id.startsWith("org.freedesktop.udisks2.")) {
      return polkit.Result.YES;
    }
  }
});
```

In the Sway config:

```
exec udiskie --tray &
```

5.3.3 Manual Unmount

```
udiskie-umount /run/media/$USER/DRIVE_NAME
udiskie-umount --all
```

5.4 MTP (Phone via USB)

```
sudo xbps-install go-mtpfs libmtp
sudo usermod -aG plugdev $USER
```

5.4.1 FUSE Setup

```
# Uncomment the user_allow_other line
sudo sed -i '/user_allow_other/s/^#//g' /etc/fuse.conf
```

5.4.2 Usage

```
# Detect connected devices
mtp-detect

# Mount the device
mkdir -p ~/phone
go-mtpfs ~/phone &

# Unmount
fusermount -u ~/phone
```

Chapter 6

Networking (VPN, Hotspot, Tethering)

6.1 NetworkManager Setup

```
sudo xbps-install NetworkManager network-manager-applet

# Stop old services
sudo sv stop dhcpcd wpa_supplicant 2>/dev/null
sudo rm /var/service/dhcpcd /var/service/wpa_supplicant
   2>/dev/null

# Enable NetworkManager
sudo ln -s /etc/sv/dbus /var/service/
sudo ln -s /etc/sv/NetworkManager /var/service/
```

6.1.1 Static DNS (optional)

```
sudo nano /etc/resolv.conf
```

```
nameserver 1.1.1.1
nameserver 8.8.8.8
```

```
sudo chattr +i /etc/resolv.conf   # Prevent NetworkManager
   from overwriting it
```

6.2 VPN

6.2.1 Install Plugins

```
# OpenVPN
sudo xbps-install NetworkManager-openvpn

# WireGuard
```

```
sudo xbps-install wireguard-tools NetworkManager-wireguard

# L2TP/IPSec
sudo xbps-install NetworkManager-l2tp
```

6.2.2 Add a VPN Connection

```
# Import a config file
nmcli connection import type openvpn file
   /path/to/config.ovpn
nmcli connection import type wireguard file
   /path/to/config.conf

# Using the TUI
nmtui
```

6.2.3 Enable / Disable

```
nmcli connection up VPN_NAME
nmcli connection down VPN_NAME
nmcli connection show
```

6.3 Hotspot

Warning: Most WiFi cards do not support internet access and hotspot simultaneously on the same adapter.

6.3.1 Quick Start

```
nmcli device wifi hotspot ifname wlan0 ssid "MyNetwork"
   password "MyPassword"
```

6.3.2 Show QR Code

```
nmcli device wifi show-password
```

6.3.3 Persistent Hotspot

```
nmcli con add type wifi ifname wlan0 con-name "MyHotspot"
   autoconnect no \
  ssid "MyNetwork" \
  802-11-wireless.mode ap \
  802-11-wireless.band bg \
  ipv4.method shared \
  wifi-sec.key-mgmt wpa-psk \
  wifi-sec.psk "MyPassword"
```

```
nmcli con up MyHotspot
nmcli con down MyHotspot
```

6.4 USB Tethering (Phone Internet on Laptop)

```
# Connect your phone via USB and enable USB Tethering on it
ip link show              # A new interface will appear
   (usb0 or enp...)
nmcli device status       # NetworkManager should detect it
   automatically
```

Chapter 7

Applications

7.1 Browser

```
sudo xbps-install firefox
```

Firefox runs natively on Wayland as long as `MOZ_ENABLE_WAYLAND=1` is set in your environment.

7.2 Multimedia

```
# Video
sudo xbps-install mpv        # native Wayland
# or
sudo xbps-install vlc

# Images
sudo xbps-install imv         # native Wayland

# Audio control
sudo xbps-install pavucontrol
```

7.3 File Manager

```
# Terminal
sudo xbps-install nnn

# Graphical
sudo xbps-install thunar
```

7.4 Archive Tools

```
sudo xbps-install unzip zip p7zip file-roller
```

7.5 System Monitoring

```
sudo xbps-install htop btop fastfetch lm_sensors
```

7.6 Flatpak

```
sudo xbps-install flatpak xdg-user-dirs
flatpak remote-add --if-not-exists flathub
   https://dl.flathub.org/repo/flathub.flatpakrepo

# Install an app
flatpak install flathub com.discordapp.Discord

# Run an app
flatpak run com.discordapp.Discord

# Update all
flatpak update

# List installed apps
flatpak list
```

7.7 AppImage

```
sudo xbps-install fuse
sudo modprobe fuse

chmod +x AppName.AppImage
./AppName.AppImage

# If it complains about FUSE
./AppName.AppImage --appimage-extract-and-run

# Organized location
mkdir -p ~/AppImages
```

7.8 Emacs

```
sudo xbps-install emacs        # with GUI
sudo xbps-install emacs-nox    # terminal only (lighter)
```

Config location: `~/.config/emacs/init.el`

7.9 Terminal Tools for Coding

```
sudo xbps-install \
  ripgrep \
  fd \
  bat \
  eza \
  fzf \
  zoxide \
  btop \
  lazygit
```

Tool	Replaces	Benefit
rg	grep	Much faster
fd	find	Simpler and faster
bat	cat	Syntax highlighting
eza	ls	Cleaner and more informative
fzf	—	Fuzzy finder for everything
zoxide	cd	Remembers visited directories
btop	htop	More features and better visuals
lazygit	git	TUI for git

7.9.1 Link fzf to Shell History

Add to `~/.bashrc`:

```
source /usr/share/fzf/key-bindings.bash
source /usr/share/fzf/completion.bash
```

Then `Ctrl+R` will open a fuzzy search over your shell history.

Chapter 8

AppImage and Desktop Entries

8.1 Directory Structure

```
~/AppImages/
 +-- obsidian.AppImage
 +-- ticktick.AppImage
 +-- icons/
     +-- obsidian.png
     +-- ticktick.png

~/.local/share/applications/    # Desktop entries
~/.config/mimeapps.list         # MIME associations
```

8.2 Setting Up a New AppImage

```
# 1. Create directories
mkdir -p ~/AppImages/icons
mkdir -p ~/.local/share/applications

# 2. Move the AppImage
mv downloaded.AppImage ~/AppImages/appname.AppImage
chmod 755 ~/AppImages/appname.AppImage

# 3. Place the icon
cp icon.png ~/AppImages/icons/appname.png
chmod 644 ~/AppImages/icons/appname.png
```

8.3 Desktop Entry Template

```
nano ~/.local/share/applications/appname.desktop
```

```
[Desktop Entry]
Version=1.0
Type=Application
```

```
Name=Application Name
Comment=Short description
Exec=/home/USERNAME/AppImages/appname.AppImage %F
Icon=/home/USERNAME/AppImages/icons/appname.png
Terminal=false
Categories=Office;Utility;
MimeType=text/markdown;text/plain;
StartupNotify=true
```

```
chmod 644 ~/.local/share/applications/appname.desktop
update-desktop-database ~/.local/share/applications
```

8.4 Exec Variables

Variable	Meaning
%f	Single file
%F	Multiple files
%u	Single URL
%U	Multiple URLs

8.5 Common MIME Types

```
text/plain        ->  .txt
text/markdown     ->  .md
text/html         ->  .html
application/pdf   ->  .pdf
image/png         ->  .png
image/jpeg        ->  .jpg
```

8.5.1 Useful Commands

```
# Get the MIME type of a file
xdg-mime query filetype document.md

# Get the default application for a type
xdg-mime query default text/markdown

# Set the default application
xdg-mime default obsidian.desktop text/markdown
```

8.6 Full Example: Obsidian

```
mkdir -p ~/AppImages/icons
mv Obsidian-*.AppImage ~/AppImages/obsidian.AppImage
```

```
chmod 755 ~/AppImages/obsidian.AppImage
cp obsidian-icon.png ~/AppImages/icons/obsidian.png
chmod 644 ~/AppImages/icons/obsidian.png
```

```
# ~/.local/share/applications/obsidian.desktop
[Desktop Entry]
Version=1.0
Type=Application
Name=Obsidian
Comment=Knowledge base and note-taking app
Exec=/home/USERNAME/AppImages/obsidian.AppImage %F
Icon=/home/USERNAME/AppImages/icons/obsidian.png
Terminal=false
Categories=Office;TextEditor;Utility;
MimeType=text/markdown;text/plain;
StartupNotify=true
```

```
chmod 644 ~/.local/share/applications/obsidian.desktop
update-desktop-database ~/.local/share/applications
xdg-mime default obsidian.desktop text/markdown
```

8.7 Web App as a Desktop Entry

```
[Desktop Entry]
Version=1.0
Type=Application
Name=Gmail
Exec=firefox --new-window https://mail.google.com
Icon=/home/USERNAME/AppImages/icons/gmail.png
Terminal=false
Categories=Network;WebBrowser;
StartupNotify=true
```

8.8 Quick Reference

```
chmod +x file.AppImage
chmod 644 file.desktop
update-desktop-database ~/.local/share/applications
xdg-mime default app.desktop text/markdown
xdg-mime query default text/markdown
gtk-launch appname              # Test a desktop entry
xdg-open file.md                # Open with the default
   application
desktop-file-validate app.desktop
```

Chapter 9

KVM/QEMU Virtualization

9.1 Verify CPU Virtualization Support

```
lscpu | grep -i virtualization
grep -E '(vmx|svm)' /proc/cpuinfo
```

- **Intel:** Must show `VT-x` or `vmx`
- **AMD:** Must show `AMD-V` or `svm`

If nothing appears, enable it in the BIOS: look for *Virtualization Technology* or *Intel VT-x*.

9.2 Installation

```
sudo xbps-install virt-manager libvirt qemu dnsmasq
   bridge-utils openbsd-netcat
```

Package	Purpose
virt-manager	GUI for managing VMs
libvirt	Virtualization API
qemu	The emulator
dnsmasq	DNS/DHCP for virtual networks
bridge-utils	Network bridging

9.3 Enable Services

```
sudo ln -s /etc/sv/dbus /var/service/        # if not
   already enabled
sudo ln -s /etc/sv/libvirtd /var/service/
sudo ln -s /etc/sv/virtlockd /var/service/
sudo ln -s /etc/sv/virtlogd /var/service/
```

9.4 Add User to the Group

```
sudo usermod -aG libvirt $USER
# Log out and back in
```

9.5 Load the KVM Module

```
# Intel
sudo modprobe kvm_intel

# AMD
sudo modprobe kvm_amd

# Load automatically on boot (Intel)
echo "kvm_intel" | sudo tee /etc/modules-load.d/kvm.conf
```

9.6 Configure libvirt (to run VMs without sudo)

```
sudo nano /etc/libvirt/libvirtd.conf
```

Uncomment these lines:

```
unix_sock_group = "libvirt"
unix_sock_ro_perms = "0777"
unix_sock_rw_perms = "0770"
```

```
sudo nano /etc/libvirt/libvirt.conf
```

Uncomment:

```
uri_default = "qemu:///system"
```

```
sudo sv restart libvirtd
sudo reboot
```

9.7 Post-Reboot Verification

```
lsmod | grep kvm
ls -la /dev/kvm
sudo virsh list --all
groups | grep libvirt
```

9.8 Essential virsh Commands

```
virsh list --all                # List all VMs
virsh start <vm-name>           # Start a VM
virsh shutdown <vm-name>        # Graceful shutdown
virsh destroy <vm-name>         # Force shutdown
virsh undefine <vm-name>        # Delete a VM
virsh dominfo <vm-name>         # Show VM info
virsh edit <vm-name>            # Edit VM config
```

9.9 Nested Virtualization

```
# Intel
sudo modprobe -r kvm_intel
sudo modprobe kvm_intel nested=1

# Persistent
echo "options kvm_intel nested=1" | sudo tee
   /etc/modprobe.d/kvm.conf
```

9.10 Troubleshooting

```
# "Unable to connect to libvirt"
sudo sv status dbus
sudo sv status libvirtd
sudo sv up libvirtd

# "Could not access KVM kernel module"
lsmod | grep kvm
sudo modprobe kvm_intel
ls -la /dev/kvm

# "Permission denied"
groups | grep libvirt
sudo usermod -aG libvirt $USER
# Log out and back in
```

Chapter 10

Troubleshooting

10.1 Sway Won't Start

10.1.1 "Failed to create backend"

```
# Verify elogind or seatd
sudo sv status elogind
sudo sv status seatd

# If using seatd, verify group membership
groups | grep _seatd

# Verify dbus
sudo sv status dbus
```

10.1.2 Sway starts and crashes immediately

```
dbus-run-session sway 2>&1 | head -30
```

10.2 XDG_RUNTIME_DIR Issues

```
# Temporary manual fix
export XDG_RUNTIME_DIR=/run/user/$(id -u)
sudo mkdir -p $XDG_RUNTIME_DIR
sudo chown $USER:$USER $XDG_RUNTIME_DIR
sudo chmod 0700 $XDG_RUNTIME_DIR

# Permanent fix: pam_rundir (with seatd)
sudo xbps-install pam_rundir
# Add to /etc/pam.d/system-login:
# -session optional pam_rundir.so
```

10.3 Audio

10.3.1 PipeWire not running

```
sudo sv status dbus
ls ~/.config/pipewire/pipewire.conf.d/
pipewire &
pactl info
grep pipewire ~/.config/sway/config
```

10.4 Firefox Not Running on Wayland

```
echo $MOZ_ENABLE_WAYLAND
# Expected: 1

MOZ_ENABLE_WAYLAND=1 firefox
```

10.5 Screen Sharing Not Working

```
# 1. Verify installation
xbps-query xdg-desktop-portal-wlr

# 2. Check portal config
cat ~/.config/xdg-desktop-portal/sway-portals.conf

# 3. Check dbus environment variables
dbus-update-activation-environment --print-environment |
   grep XDG_CURRENT_DESKTOP

# 4. Restart portals manually
killall xdg-desktop-portal xdg-desktop-portal-wlr
   2>/dev/null
/usr/libexec/xdg-desktop-portal-wlr &
/usr/libexec/xdg-desktop-portal -r &

# 5. Test in Firefox
# https://mozilla.github.io/webrtc-landing/gum_test.html
```

10.6 Qt Apps Showing as X11

```
echo $QT_QPA_PLATFORM
# Expected: wayland

sudo xbps-install qt5-wayland qt6-wayland
```

10.7 Java Apps (Blank Grey Window)

```
# Add to ~/.profile
export _JAVA_AWT_WM_NONREPARENTING=1
```

10.8 AppImage Won't Launch

```
sudo xbps-install fuse
sudo modprobe fuse
chmod 755 ~/AppImages/app.AppImage
./app.AppImage --appimage-extract-and-run
```

10.9 General Diagnostic Commands

```
echo $WAYLAND_DISPLAY        # Expected: wayland-1
swaymsg -t get_version
swaymsg -t get_outputs
swaymsg -t get_inputs
dbus-update-activation-environment --print-environment
```

Chapter 11

xbps-src: Building Packages Not in the Repo

11.1 The Core Concept

`xbps-src` is similar to `PKGBUILD` on Arch Linux or `ebuild` on Gentoo.

Instead of building from source manually, you write (or copy) a small "recipe" file called a **template**. Then `xbps-src` builds the program and creates an `.xbps` package that you can install and remove with `xbps` like any other package.

> **The main benefit:** The program becomes managed — you can remove it with `xbps-remove` just like any other package.

11.2 Initial Setup (One Time Only)

```
sudo xbps-install git xtools base-devel

cd ~
git clone --depth=1
   https://github.com/void-linux/void-packages.git
cd void-packages

./xbps-src binary-bootstrap
```

> `-depth=1` fetches only the latest snapshot without the full history — much faster.

11.3 Building a Package Already in void-packages

```
# Search for the template
ls srcpkgs/ | grep -i package_name
./xbps-src search package_name

# Build
./xbps-src pkg package_name
```

```
# Install
sudo xbps-install --repository=hostdir/binpkgs package_name

# If in nonfree
sudo xbps-install --repository=hostdir/binpkgs/nonfree
   package_name
```

11.3.1 Practical Example: Google Chrome

```
cd ~/void-packages
echo "XBPS_ALLOW_RESTRICTED=yes" >> etc/conf
./xbps-src pkg google-chrome
sudo xbps-install --repository=hostdir/binpkgs/nonfree
   google-chrome
```

11.4 Adding the Local Repo Permanently

```
sudo nano /etc/xbps.d/local-repo.conf
```

```
repository=/home/YOUR_USERNAME/void-packages/hostdir/binpkgs
repository=/home/YOUR_USERNAME/void-packages/hostdir/binpkgs/nonfree
```

11.5 Anatomy of a Template

```
pkgname=my-tool
version=1.2.3
revision=1
short_desc="An awesome tool"
maintainer="Your Name <email@example.com>"
license="MIT"
homepage="https://github.com/someone/my-tool"

distfiles="https://github.com/someone/my-tool/archive/v${version}.tar.gz"
checksum="sha256 hash goes here"

hostmakedepends="python3"
depends="python3"

do_install() {
    python3 setup.py install --prefix=/usr
       --root="${DESTDIR}"
}
```

11.5.1 Key Variables

Variable	Meaning
pkgname	Package name
version	Version number
revision	Revision number (start at 1)
short_desc	Short description (max 72 characters)
distfiles	Source URL (tar.gz or zip)
checksum	SHA256 hash of the source file
hostmakedepends	Build-time tools
makedepends	Build-time libraries
depends	Runtime dependencies
build_style	Automatic build method

11.5.2 build_style Options

```
build_style=cmake           # CMake projects
build_style=meson           # Meson projects
build_style=gnu-configure   # Autotools (./configure)
build_style=python3-module  # Python (setup.py)
build_style=go              # Go
build_style=cargo           # Rust
build_style=gnu-makefile    # Simple Makefile projects
```

11.6 Getting the Checksum

```
# Using xtools
xgensum -i srcpkgs/package_name/template

# Manually
wget https://source_url
sha256sum filename.tar.gz
```

11.7 Full Example: A Rust Package (zoxide)

```
./xbps-src newpkg zoxide
```

```
# srcpkgs/zoxide/template
pkgname=zoxide
version=0.9.4
revision=1
build_style=cargo
short_desc="Smarter cd command"
maintainer="Me <me@example.com>"
```

```
license="MIT"
homepage="https://github.com/ajeetdsouza/zoxide"
distfiles="https://github.com/ajeetdsouza/zoxide/archive/v${version}.tar.g
checksum="put sha256 here"
```

```
xgensum -i srcpkgs/zoxide/template
./xbps-src pkg zoxide
sudo xbps-install --repository=hostdir/binpkgs zoxide
```

11.8 Everyday xbps-src Commands

```
./xbps-src pkg package_name          # Build
./xbps-src -f pkg package_name       # Force rebuild from
   scratch
./xbps-src clean package_name        # Clean build files
./xbps-src show package_name         # Show dependencies
cd ~/void-packages && git pull        # Update void-packages
./xbps-src update-sys                 # Update outdated
   packages
```

11.9 Updating a Package

```
cd ~/void-packages
git pull
nano srcpkgs/package_name/template   # Change version=
xgensum -i srcpkgs/package_name/template
./xbps-src pkg package_name
sudo xbps-install -u package_name
```

11.10 Hold: Prevent a Package from Updating

```
sudo xbps-pkgdb -m hold package_name
sudo xbps-pkgdb -m unhold package_name
```

11.11 Troubleshooting xbps-src

"checksum mismatch":

```
xgensum -i srcpkgs/package_name/template
```

Build is very slow:

```
echo "XBPS_MAKEJOBS=$(nproc)" >> etc/conf
```

View the full build log:

```
./xbps-src pkg package_name 2>&1 | tee /tmp/build.log
less /tmp/build.log
```

11.12 Quick Summary

```
# Initial setup (one time)
git clone --depth=1
   https://github.com/void-linux/void-packages.git
cd void-packages
./xbps-src binary-bootstrap

# Build and install an existing package
./xbps-src pkg package_name
sudo xbps-install --repository=hostdir/binpkgs package_name

# Create a new package
./xbps-src newpkg package_name
# Edit the template
xgensum -i srcpkgs/package_name/template
./xbps-src pkg package_name
sudo xbps-install --repository=hostdir/binpkgs package_name

# Remove
sudo xbps-remove package_name
```

Chapter 12

i3 on Void Linux (Migration from Sway + Running Both)

12.1 The Core Difference in One Sentence

	Sway	**i3**
Display Server	Wayland	X11 / Xorg
Config syntax	~99% identical	~99% identical
Compositor	Built-in	Requires separate `picom`
Screen Sharing	Via Portal + PipeWire	Direct (easier)

12.2 Full Migration Table: Swap These for Those

Role	Sway (Wayland)	i3 (X11)
Window Manager	sway	i3
Status Bar	waybar	polybar or i3status
Notifications	mako	dunst
Launcher	wofi / fuzzel	rofi / dmenu
Lock Screen	swaylock	i3lock
Idle	swayidle	xss-lock + xautolock
Wallpaper	swaybg	feh / nitrogen
Screenshot	grim + slurp	flameshot / scrot
Screen Recording	wf-recorder	OBS / ffmpeg
Clipboard	wl-clipboard + clipman	xclip / xdotool
Compositor	Built into Sway	picom
Display Config	kanshi / wdisplays	arandr / autorandr
Brightness	brightnessctl (same)	
Volume	pactl (same)	
Theming	nwg-look	lxappearance
Polkit Agent	polkit-gnome (same)	
Keyring	gnome-keyring (same)	
Network Applet	nm-applet (same)	
Bluetooth Applet	blueman-applet (same)	

Items marked **(same)** do not need to be changed at all.

12.3 Installing i3 on Void

```
# i3 and X11
sudo xbps-install i3 xorg-minimal xinit

# Tool replacements
sudo xbps-install \
  polybar \
  dunst \
  i3lock \
  feh \
  picom \
  flameshot \
  xclip \
  arandr \
  lxappearance \
  xss-lock \
```

```
  xautolock
```

12.4 Launching i3 from the TTY

Create `~/.xinitrc`:

```
#!/bin/sh

# dbus must be first -- without it, PipeWire and Keyring
   won't work
eval $(dbus-launch --sh-syntax)
export DBUS_SESSION_BUS_ADDRESS

# X11 variables
export XDG_SESSION_TYPE=x11
export XDG_SESSION_DESKTOP=i3
export XDG_CURRENT_DESKTOP=i3

# Keyring (after dbus)
eval $(gnome-keyring-daemon --start
   --components=gpg,pkcs11,secrets,ssh \
    | sed 's/^\(.*\)/export \1/g')
export SSH_AUTH_SOCK

# PipeWire (after dbus)
pipewire &

exec i3
```

```
chmod +x ~/.xinitrc
```

To launch:

```
startx
```

Common mistake: If you start PipeWire before dbus, or launch it from the i3 config instead of `.xinitrc`, audio will not work. The correct order is: `dbus` → `keyring` → `pipewire` → `i3`

12.5 Auto-start from TTY1 (Sway) or TTY2 (i3)

In `~/.profile`:

```
# TTY1 -> Sway
if [ -z "$WAYLAND_DISPLAY" ] && [ "$(tty)" = "/dev/tty1" ];
   then
    exec ~/.local/bin/start-sway
fi
```

```
# TTY2 -> i3
if [ -z "$DISPLAY" ] && [ "$(tty)" = "/dev/tty2" ]; then
    exec startx
fi
```

Now: `tty1` = Sway / `tty2` = i3. Switch between them with `Ctrl+Alt+F1` and `Ctrl+Alt+F2`.

12.6 i3 Config (A Full Starting Point)

```
mkdir -p ~/.config/i3
cp /etc/i3/config ~/.config/i3/config 2>/dev/null || \
  cp /usr/share/doc/i3/config ~/.config/i3/config
     2>/dev/null || \
  i3-config-wizard
```

Add the following to `~/.config/i3/config`:

```
### Startup ###
exec --no-startup-id dbus-update-activation-environment
   --all

exec --no-startup-id pipewire &
exec --no-startup-id dunst &
exec --no-startup-id picom -b &
exec --no-startup-id feh --bg-fill ~/wallpaper.jpg &
exec --no-startup-id nm-applet &
exec --no-startup-id blueman-applet &
exec --no-startup-id udiskie --tray &
exec --no-startup-id
   /usr/lib/polkit-gnome/polkit-gnome-authentication-agent-1
   &
exec --no-startup-id xss-lock -- i3lock -f -c 000000 &

### Status Bar ###
bar {
    status_command i3status
    # or if you installed polybar:
    # exec_always --no-startup-id polybar
}

### Screenshot ###
bindsym Print exec flameshot gui
bindsym Shift+Print exec flameshot full -p ~/Pictures/

### Volume ###
bindsym XF86AudioRaiseVolume exec pactl set-sink-volume
   @DEFAULT_SINK@ +5%
bindsym XF86AudioLowerVolume exec pactl set-sink-volume
   @DEFAULT_SINK@ -5%
```

```
bindsym XF86AudioMute exec pactl set-sink-mute
   @DEFAULT_SINK@ toggle

### Brightness ###
bindsym XF86MonBrightnessUp exec brightnessctl set +10%
bindsym XF86MonBrightnessDown exec brightnessctl set 10%-

### Lock ###
bindsym $mod+l exec i3lock -f -c 000000
```

12.7 Sharing Config Between i3 and Sway (The Smart Way)

The two configs are nearly identical — you can maintain one shared config with separate override files for each WM.

12.7.1 Structure

```
~/.config/i3/config              <- Shared config
   (keybindings, layouts, etc.)
~/.config/i3/conf.d/i3.conf     <- i3-only settings
   (polybar, picom, i3lock)
~/.config/sway/config            <- Symlink to the shared
   config
~/.config/sway/conf.d/sway.conf <- Sway-only settings
   (waybar, swaylock)
```

12.7.2 Implementation

```
# Create the symlink
mkdir -p ~/.config/sway/conf.d ~/.config/i3/conf.d
ln -s ~/.config/i3/config ~/.config/sway/config
```

Add at the end of `~/.config/i3/config`:

```
# Load WM-specific extras
include conf.d/*.conf
```

In `~/.config/i3/conf.d/i3.conf`:

```
exec --no-startup-id picom -b &
exec --no-startup-id polybar &
exec --no-startup-id feh --bg-fill ~/wallpaper.jpg &
bindsym $mod+l exec i3lock -f -c 000000
```

In `~/.config/sway/conf.d/sway.conf`:

```
exec waybar &
exec swaybg -i ~/wallpaper.jpg -m fill &
```

```
bindsym $mod+l exec swaylock -f -c 000000
output * bg ~/wallpaper.jpg fill
```

12.8 Theming in i3

i3 uses `lxappearance` instead of `nwg-look`:

```
lxappearance
```

Same idea — choose GTK theme + Icons + Cursor and click Apply. On X11 it works more straightforwardly without the `gsettings` complexity.

12.9 picom (Compositor for i3)

Without it you will not have transparency or visual effects. Minimal setup:

```
# Launch from i3 config
exec --no-startup-id picom -b

# Or with a custom config
mkdir -p ~/.config/picom
nano ~/.config/picom/picom.conf
```

```
# picom.conf -- minimal setup
backend = "glx";
vsync = true;
shadow = false;
fading = true;
fade-in-step = 0.05;
fade-out-step = 0.05;

# Transparency for terminal emulators
opacity-rule = [
    "90:class_g = 'tilix'",
    "90:class_g = 'Alacritty'"
];
```

12.10 Summary: What to Install on Top of What You Have

```
# X11 base
sudo xbps-install xorg-minimal xinit

# i3 ecosystem
sudo xbps-install i3 polybar dunst i3lock feh picom
   flameshot \
  xclip arandr lxappearance xss-lock
```

```
# No need to reinstall:
# pipewire, nm-applet, blueman, udiskie, polkit-gnome,
   brightnessctl
```

12.11 Quick Verification

```
# Verify Xorg
which startx
ls /usr/bin/Xorg

# Verify i3
which i3
i3 --version

# Launch
startx
```